

OVERALL QC INTERPRETATION

Output	Result
PCA plot	The PCA plot shows the separation of samples based on their gene-expression profiles. The Infected and Control samples form separate groups, with PC1 explaining 96% of the variance and PC2 explaining 3%.
Sample-to-sample distance	This plot shows the similarity or difference in gene-expression patterns between samples. Smaller distances indicate more similar samples, while larger distances indicate greater differences.
Dispersion estimates	The dispersion plot shows the estimated variability of gene counts relative to the mean of normalized counts. It includes individual gene estimates, the fitted trend, and final dispersion values.
p-value histogram	The histogram displays the distribution of p-values for the Control vs. Infected comparison. It shows the frequency of genes across p-value ranges from 0 to 1.
MA-plot	The MA-plot shows the relationship between the mean of normalized counts and log fold change, allowing genes with increased or decreased expression to be visualized.